Distinct element analysis of underground voids in stratified rock masses

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The study proposed analyses the stress and strain state induced by the excavation of underground voids in fractured rock masses, in particular those of a stratified type. This is an interesting and widespread problem. Stability, in these cases, can be put down to the self-support of the roof layer, whose most reliable modelling is that of the "voussoir beam" (jointed rock beam) analogue. The analyses that were carried out have identified the key parameters that influence the equilibrium of the system. The possibility of defining the trend of the excavation span under critical stability conditions through numerical means allowed a comparison with the design criteria that are available in literature; the obtained results allow one to understand with what degree of approximation such criteria can deal with the real behaviour of the system and to evaluate the applicability at a design level. After a study of the behaviour of a single layer, one was carried out on multilayered voussoir beams systems, which simulate the entire overburden of the excavation; the validation of the approach to the problem of stability using the voussoir beam analogue led to the conclusion that the application of underground excavation design criteria obtained from literature is cautionary. Finally, significant comparisons are shown that allow one to evaluate the degree of approximation of the behaviour of tunnels, from these simple models that were introduced.

1. Introduction

The structural particularities of a rock mass that can be encountered during an underground excavation can be varied and very different from each other, making the choice of the most adequate project procedure that is able to approximate the real behaviour of vital importance. Some of the most widely used theories in the ambit of the designing of underground openings in fractured masses are presented in this work; the limits of applicability are explained and the validity fields are justified through the numerical study of the particular mechanisms that guarantee stability and that take place in the rock mass following excavation.
The so-called *"disturbed zone"* inside an excavation, in stratified masses, is governed by a redistribution and mechanism of stresses transfer, which lead to the formation of a zone of "loose" material above the excavation which, undergoing tensile stresses, tends to break off from the surrounding rock due to the modest values in the ratio between tensile and compression strength along the joints. The system would rapidly become unstable if it were not for the development of a compression stress state in the layer that makes up the roof above, which is called *"linear arch"* or *"roof-arch"*; in other words the stability of the excavation is connected to the self-support of the roof layer which is usually analysed through the most adherent modelling of the real conditions: the *"voussoir beam"* model.
The stability of the roof layer is a function of numerous parameters, which can be put down to four fundamental ones. Among these, the role played by the thickness of the beam allows to define, through the use of numerical models, the trend of the excavation span in critical stability conditions in function of the mean thickness of the stratification, thus making a comparison possible with design criteria found in literature.
Although the model of the single *voussoir beam* constitutes a valid design instrument, *multilayered beams systems* better capture the real behaviour of the entire excavation overburden. The possibility of considering fundamental factors for the stability in such models, such as inter-layer friction, allows one to both verify the redistribution of the stresses and the consequent diversification of the static mechanisms, due to the occurrence of the arch effect and also to characterise the static behaviour of the two ideal regions that make up the excavation overburden, which are subject to typical occurrences of the arch effect: *ground and roof-arch effects.*
Finally, comparisons are presented between elementary and numerical tunnel models in order to verify their capacity to collect the induced stress and strain aspects, especially in the zones that are the most significant for stability.

2. The "disturbed zone"

The *"disturbed zone"* [3] around an excavation is the zone where the original state of the rock mass, in terms of *stress, deformation, stability* and *water flow states*, is modified by the construction of an underground work. Inside it is possible to identify three ideal sub-zones, which are different because of their peculiarities and engineering importance.

- the *collapse* zone, where instability phenomena can occur during or immediately after excavation. Free blocks fall, because of the removal of the excavation material, and it is necessary to intervene immediately with local reinforcement;
- the *open joints* zone, is characterised by opening or loosening of the joints and long term instability can arise or a significant increase in the water flow can occur;
- the *shear* zone, which contains the region of the model in which the joints undergo the greatest sliding. It is important in that, on one hand, it is extremely dangerous for the stability of the tunnel due to the risk of shear rupture of the reinforce-

ment elements while, on the other, the sliding of the joints can determine preferential paths for the water flows, due to the dilatancy effect.

Among the various theoretical approaches, *"Distinct Element"* modeling is that which allows one to consider the different aspects in the three zones in detail to the greatest extent. Using numerical models (UDEC, Itasca 1996), numerous analyses have been carried out that simulate the work of a deep excavation in rock, varying the geometric parameters (spacing and orientation of the joints) and the mechanical parameters (friction mobilised along the discontinuity and original stress state).

The analysed models refer to a tunnel situated at 100 m in depth with a diameter of 10 m. The rock material, with the mechanical characteristics that are reported in tab.1, was considered to be elastic, while the joints were considered to be elastic-plastic, according to the Mohr-Coulomb resistance criterion. The two families of orthogonal joints show the same mechanical characteristics, the spacings vary from 4 to 1 m and the number of blocks vary from 2500 to 10,000.

The main conclusions that can be drawn from the analysis of the results are as follows:

- the collapse zone is always limited to the surfaces of the excavation. Its extension is strictly connected to the spacing of the discontinuities and their dips. The effects induced by the variation of the friction angle and of the original stress state appear to be modest;
- the open joints zone becomes larger with a diminishing of the spacings of the joints and with a decrease in the severity of the original stress state. For the minimum values that the thrust coefficient at rest Ko can assume in rock masses (Ko = 0.25), it shows an extension that is no greater than the tunnel radius. Values of Ko slightly higher than 0.5 cm reduce it so much as to almost eliminate it completely;
- the shear zone is very sensitive to the entity of the spacings of the joints and to the characteristics of the original stress state; modest spacings and low values of the thrust coefficient at rest can provoke a considerable increase, reaching a distances from the walls of 10 times the radius. Other situations that particularly favour the extension are the concomitance of modest friction angles and of dips of the discontinuities around 60°.

An example of the typical configuration assumed by the three zones, with variations of the dips of the discontinuities, is shown in figure 2.1 (cases Jdip = 60°, 45°).

The analysis of the three zones and the configuration of the principal stresses at equilibrium allow to unequivocally recognise that the manifestations of the arch effect bring the stability of the excavation back to that of the roof layer, whatever the dip of the main discontinuities.

3. The voussoir beam

During excavation of a void it is quite improbable that a roof made up of thin layers could be completely continuous; due to the effect of a progressive fracturing phenomenon, the transition from an original behaviour of an *elastic continuous beam* type (characterised by closed form analytical solutions) to a *voussoir beam* type (in which the bending and stability problem is statically undetermined) is plausible. The analytical solutions presented in literature supply the *thickness of the compressed zone (N·T)* and the *maximum (horizontal) compression stress* (σ_{max}) so as to be able to calculate, knowing the uniaxial compression strength of the rock and the shear resistance along the joints, the safety coefficient against the collapse modality of a bending instability type (buckling), localised compression (crushing) or shear sliding along the extremities (sliding), or, once the later is imposed (Fs), the span that guarantees stability conditions thanks to the setting up of the roof arch.

3.1 Bending and stability of the voussoir beams

The stability conditions of the voussoir beams depend not only on the *geometric and mechanical magnitudes* (resistance and deformability parameters of the material and of the joints), but also on the *initially imposed stress state* (confinement).

Having chosen the *maximum deflection* (f) as the representative parameter of the bending phenomenon, it is possible to represent it, in most cases, with the following expression:

$$f = f\,[a,\ T,\ S,\ Kn,\ Ks,\ \phi\ (o\ tg\phi),\ \gamma,\ E\ ,\sigma_o] \qquad (1)$$

where:

a = longitudinal dimension of the blocks;
T = thickness of the blocks;
S = span of the beams;
Kn = normal stiffness of the joints;
Ks = tangential stiffness of the joints;
$tg\phi$ = friction along the joints;
γ = weight of the volume unit;
E = elastic modulus of the rock;
σ_0 = lateral confinement.

A complete analytical treatment of relation (1) is only possible in particularly simple cases. More often a solution can be obtained experimentally or through the use of numerical models, which are particularly complex because of the high number of variables at stake. The possibility of reducing their number was therefore verified, and those that were determinant for the stability of the system were identified.

As both the normal stiffness of the joints (Kn), their frequency (n° of joints/metre), connected to the a/T ratio, and the elastic modulus of the rock (E) all influence the longitudinal deformability of the entire beam, we asked ourselves if it were possible to bring the overall behaviour of these parameters to their *global equivalent stiffness* function (Req).

The value of γ constant and the contribution of the bending supplied by the tangential stiffness (Ks), which is negligible compared to that due to the combined effect of the friction angle available along the joints and of the confinement applied from the outside, were also presumed. The geometry of the beam, in particular that of the thickness (T), (which in reality represents the mean thickness of the stratification), plays an important role on the buckling type instability phenomena.

On the basis of these considerations it is possible to connect relation (1) to:

$$f = f\,[Req,\ \sigma_o,\ \phi,\ T] \qquad (2)$$

with a noteworthy simplification of the numerical methods.

3.2 Effect of the equivalent deformability

Numerical analyses were performed in order to evaluate the conditions in which it is possible to represent the behaviour of the voussoir beam with the global equivalent stiffness parameter (Req), defined by the function:

$$\frac{1}{Req}=\frac{1}{E_{roccia}}+\frac{1}{Kn}\cdot\frac{n}{S} \qquad (3)$$

(where n = the number of joints and S = the beam span).

The study hypothesises a span beam S = 12 m and a thickness T = 1 m. The blocks are elastic and the joints elastic-plastic according to the Mohr-Coulomb criteria.

The effect of the elastic modulus E is particularly evident only for values lower than 10 GPa, the interval in which, with the geometry and conditions considered, differences in the maximum deflection are registered even up to 100% compared with the value corresponding to the case E = 10 GPa. For higher values of E, which is typical of consistent rocks in which the "roof arch" effect develops more frequently, this dependence is lower, independently of the confinement value.

The equivalent longitudinal stiffness Req was therefore correlated to the maximum deflection of the beams (f), varying the frequency of the joints (a/T), of the value Kn and of the elastic modulus of the rock (E), with the conditions being the same and for different values of applied confinement ($\sigma o/\gamma T$= *2, 0.4, 0.04*).

From figure 3.1 it can be seen that, for high values of Req (Req > 12 GPa) and of the confinement, the dependence seems so pronounced that the obtained curves result to be coincident, regardless of which parameter is varied (Kn, E or a/T). For mean values of Req (0.8GPa < Req <12GPa) the dependence is obvious, especially for medium and high confinement values; in such situations differences are encountered in the maximum deflection values which, in percentage terms, are around 37%. With a diminishing of the confinement (or even of the friction), the effect of the individual parameters is more obvious. For low values of Req (Req < 0.8GPa) and of the confinement, the beam becomes unstable.

3.3 The effect of the confinement and friction parameters

With the some beam geometry, the adopted mechanical properties are reported in table 1, where ϕ varies between 14° and 60°. Three geometries have been assumed for the blocks, which differ according to the values of the dimensions *a* : 2.0, 1.5 and 0.75 m. Three values were also assumed for the confinement.

The effect on the *deformation behaviour* of the system is synthesised in the graph of fig. 3.2.

It is possible to identify a *threshold* in the *friction* values (situated around 40°), above which the effect of the confinement on the deflection of the beam is not particularly appreciable. With a diminishing of the confinement, the progressive reduction of the friction produces increases in the arrows, even up to values four times those in correspondence to the threshold itself. Finally, the diminishing of the friction to the lower values of the confinement favours *anticipated* collapses from *large sized deformations*, reducing the tendency to collapse of a *sudden* type.

The effects of the two parameters on the *stability conditions* of a voussoir beam is summarised in the graph of fig. 3.3 which shows, for each of the block geometries and for each of the imposed confinement values, the critical values of ϕ (ϕ_{crit}), above which the collapse of the system occurs. The progressive increase of the frequency of the fracturing determines an extension of the potential instability zone of the system. The study also shows the existence of a *limit value of the critical friction angle* of the joints (12°-14°), below which stability conditions are not encountered, regardless of how high the confinement values imposed from the outside are.

3.4 The influence of the thickness on the bending of a voussoir beam

The *mean thickness of the stratification* (T) plays a key role in the stability of the roof of an excavation. The presented numerical study identifies curves that limit the excavation span, in *limit collapse situations*, in function of the thickness of the layer (T) and compares it with those that can be obtained from analytical models obtained from literature and which are normally used in the design environment, identifying which are the most suitable design criteria that are able to deal with the real phenomenon from among the various design criteria.

a) The models and design criteria

The analytical models and the design criteria that are available in literature divide voussoir beams into: *thick beams*, characterised by low values of the span-thickness ratio (S/T) and by rigid displacements of the blocks, and *thin beams*, characterised by a relevant effect of the deformability of the blocks and by maximum deflection of entities that can be compared to the thickness. A different approach to the study corresponds to each type: equilibrium conditions imposed with respect to the initial configuration of the beam for the first case and respect of the deformed configuration in the second case. It is also possible to identify *traditional models* (Evans, 1941; Cox, 1974 [6]; Beer & Meek, 1982 [4]), which presume that both the mean thickness of the arch that originates in the beam and the distribution of the stresses in its critical sections (abutments and midspan) are a priore defined; *iterative type models* (Brady and Brown, 1993 [5]; Diederichs and Kaiser (D &K), 1999 [7]), which presume only the (parabolic) trend of the stresses within the beam are a priore defined but consider the extension of the reactive section at the abutments and at midspan to be unknown.

The comparison basically involves the D.&K. iterative method, as it is the most recent. We have introduced the *buckling limit* (B.L.) from their study [7] to be used as follows. The method consists of an iterative procedure that, minimising the *maximum compression stress (fmax)*, identifies two magnitudes previously unknown at the equilibrium: the thickness of the arch ($N \cdot T$) (reactive section) in the critical sections of the model and the *value of the arm of the stabilising couple Z* (Z = f (N). With a variation of N between 0 and 1, it is always possible, for a stable beam (span lower than the critical one), to identify values of Z: in this case the equilibrium corresponds to the minimum value fmax. Approaching the *critical span*, the percentage of the values of N that lead to possible values of Z di-

minishes. The percentage of the values of N for which no solution exists is defined as the *Buckling limit* (B.L.). The B.L. parameter therefore identifies the probability with which the beam could collapse according to this modality.
Studies on beams of different stiffnesses and thicknesses have led the authors to identify a design criterion based on the limitation of the displacements (f). D. & K. identify a B.L. of 100% at the ultimate collapse, which corresponds to a value of the f/T ratio = 0.25. The beginning of the non linearity in the f T connection corresponds to a B. L. = 35% to which a value of the f/T ratio = 0.1 corresponds. The choice, during the design stage, of the buckling limit (B.L.) directly influences the critical span, once the thickness has been fixed, or, vice versa, the value of the critical thickness, whenever the face is fixed. In the measurement stage, a limit condition for the maximum deflection equal to 10% of the thickness of the layer (B.L. = 35%), results to be conservative. Comparisons between *critical excavation spans* obtained through analytical methods in the cases of thick and thin beams, and under the different hypotheses of distribution of the stresses, have shown that, at low values of the thickness, the hypothesis of thick beam leads to an overestimation of the real critical span. In the numerical study it is therefore reasonable to adopt the hypothesis of thin beams, *whatever the reality under examination*, as it is more cautionary.

b) Particular aspects of the comparison between numerical and analytical models

The study, carried out with a distinct element method (UDEC), has examined beams subject to the action of their own weight, with a span equal to 8, 12, 24, 36 and 48 m, characterised by blocks of a width of 2 m. The properties of the rock material and of the joints are shown in table 2.
The numerical simulation refers to thin blocks and *buckling* type instability phenomena, often known as *snap-through*. Other phenomena have been neglected, e.g. *sliding*, as this is typical of thick beams and *crushing*, as this concerns both types of beams but occurs more frequently in thicker beams. Comparison between numerical and analytical models are possible through the magnitude Req (3).
Some of the most significant aspects that emerged from the analysis and from the comparison are here reported.
Correlating the *thickness* (T) values of the beam in imminent *buckling* conditions with its span (fig. 3.4) it is possible to distinguish stable and unstable geometric configurations in the span-thickness plane. Curves 2 and 3 refer to the application of the D.& K. algorithm in imminent *buckling* conditions (B.L. = 90%) and in adequately safe conditions of such a phenomenon (B.L. = 35%), respectively. The trend of the curves, which are almost identical to that obtained numerically (curve 1), confers validity to the method in terms of representativness of the phenomenon; the position on the right of curve 2 (B.L. = 90%) compared to 1 (numerical study), shows a non conservative estimation of the critical stability conditions: the mean difference between the values supplied by the curves is 13%. The adoption of a B.L. = 35% determines the displacement of curve 2 to the left of curve 1, allowing a conservative estimation of the excavation span and demonstrating the validity of application of the method for design purposes.
As far as the aspects that are more closely connected to the mechanism at the base of the equilibrium of the beam (roof arch) are concerned, from the numerical analysis it can be observed that:

- at the critical collapse limit, independently of the width of the excavation span (S), the reactive section of the beam is equal to 45% of its thickness ($N_{average}$ = 0.45) at midspan, while it is equal to about 20% ($N_{average}$ = 0.19) at the abutments. The value of N that can be obtained with the D.& K. method, and under the same conditions, is around 33%. The values adopted by the other authors, in particular by Brady and Brown (N = 0.75) and Evans (N = 0.5) do not agree with the numerical results; on the basis of some considerations made by Sterling (1980), this can be considered to be due to an increase in the area of contact between the blocks, whenever the uniaxial compression strength of the rock is reached before *buckling* is triggered.
- The maximum longitudinal compression stresses (σ_{max}), at the abutments and at midspan, increase almost linearly with the span. The different extension of the reacting sections however determine a maximum stress at the abutments which is on average 20% greater than at midspan. The trends that can be obtained with the D.& K. method, which considers the maximum stresses in the two critical sections to be equal, are close to those obtained with UDEC, especially in the case of the higher span.
- The aspects connected to the value of the normalised arm (Z/T) and the maximum deflection of the system (f) under stability limit conditions due to buckling are of significant importance.

The incipient collapse, the mean value of the arm of the thrust couple is constant and equal to 70% of the thickness of the beam (Z/T = 0.7), independently of the width of the span. The D.& K. method supplies a mean value equal to 50% (Z/T = 0.5). Under the same conditions, the mean value of the numerically obtained maximum deflection (f) is equal to 22% of the thickness of the beam. The analogous value obtained through analytical means is equal to 25%.
The substantial agreement of the results confirm, on one hand, the adequacy of the method to take buckling type instability phenomena into account, and on the other hand, the validity of the design criteria proposed by the authors (limiting of the displacements).

c) Comparison between the numerical results and analytical methods

The span-thickness curves relative to the solutions for thin beams obtained through application of the hypotheses proposed by Evans (1941), Cox (1974), Brady and Brown (1985), and Diederichs and Kaiser (1999), are given in figure 3.4. For the determination of the curves of the first three authors, the uniaxial compression strength (σ_0) of the materials close to the joints is considered to be known a priori. The mechanical parameters that are used are those that are shown in table 1. The geometric parameters of the beam are: S = 12 m, a = 2 m and T = 1 m. The curves therefore represent the *design limit curves* for collapse modalities of the localised compression type (crushing), with the exception of curve 2

(D.& K., B.L. = 90%), in which the uniaxial compression strength value is not included in the algorithm. The curve 3 has not been considered since it's not representative of critical conditions.
The graph of figure 3.4 shows that, the mean thickness of the layers being known, the critical span of the excavation values obtained through classical methods result to be *more conservative* than those obtained through the numerical analyses. Among these, the most cautionary of all is that which is based on the stress distribution by Cox. It is worthwhile underlining the conceptual differences in the type of collapse the curves refer to: *crushing* for analytical methods, *snap-through* for numerical models and for the analytical methods of D.& K.
From a comparison of the analytical and numerical results, in the particular considered case of uniaxial compression strength close to the joints, which is typical of altered joints equal to 20 MPa, it can be seen how the collapse is in fact of a *crushing* type only for large span openings; for modest spans, the collapse that first occurs is that due to *buckling* (fig. 3.5).

4. The multilayered voussoir beams systems

Span beam models have been hypothesised with S = 12 m and thickness T = 1 m. The number of the layers (n) varies from 2 to 70. The block form factor a/T (width/height) assumes the values 6, 2, 1 and 0.6. The behaviour models adopted for the rock material and for the joints are the same ones that were used in the previous analyses. The adopted mechanical parameters are those that are shown in table. 1 with $\phi=50°$.
Examining the distribution of the principal stresses, it is possible to note that, in the models in which the representative point is before *the peak* (fig. 4.1 left side), the layers show an *independent* behaviour (fig. 4.3), highlighted by the parabolic trend of the principal stresses in the *single layer*; in the models that occur *after the peak* (fig. 4.1 right side), a redistribution of the stresses can be seen (fig. 4.8) that, in the overburden of the void, causes a *higher zone* in which the trend of the maximum principal stresses follow an approximately parabolic directrix that involves a remarkable number of layers; and a *lower zone* in which the stresses again follow a parabolic direction within the single layer that tends to behave in an independent way and to be subject to its own weight (fig. 4.4). This zone is subject to the so-called *"border effect"* and on the inside the layers tend to separate one from the other (fig. 4.2).
In short, at *modest depths* (fig. 4.1 left side), the trend of the principal stresses, independently of the spacings of the joints, leads the stability of the excavation to manifest the so-called *roof-arch* effect; at large depths (fig. 4.1, right part), the diversification of the static mechanisms leads the stability to the contemporary presence of a *ground-arch* effect and a *roof-arch* effect in the roof immediately above.
The behaviour of independent beams, which is typical of shallow excavation overburdens, is characterised not only by an equal trend of the vertical stresses σ_y and horizontal stresses σ_x within the blocks of each layer, but above all by the absence of normal and tangential stresses along the layer planes (fig. 4.5 4.6). In deep excavations, an evident collaboration between layers in the ground-arch zone, which makes the model behave like a single means of a *continuous widely extended* type, is accompanied by the contemporary manifestation of both mechanisms. This collaboration is evident not only from the trend of the vertical stresses σ_y in the blocks, but above all from the non null values of the normal (fig. 4.7) and tangential stresses (fig. 4.8) to the layer planes.
With an increase in the depth of the excavation, it is possible to identify a limit value, which is a function of the spacing, over which the extension of the area subjected to important *border effects* does not undergo any further modifications (fig. 4.2). Furthermore, whatever the depth of the excavation, the entire part of the model that proceeds the *"border effect"* zone is subject to *ground-arching*. It follows that the excavation of a void at a great depth gives rise, from a *stress-strain* point of view, to a less serious condition than one could expect: only a small area of the rock mass, that in fact defined as having *"border effects"* and whose extension is inversely proportional to the frequency of the fracturing, influences its stability (fig. 4.2).
We here illustrate what has been so far explained with two examples.
A model (n = 21) is shown in figure 4.3 that, in the case of a 2 m spacing, simulates a shallow excavation. The entire model is subject to "border effects": the beams have independent behaviour and are only subject to their own weight; the stresses that are *normal* and *tangential* to the layer planes are null along almost the whole length of their extension (figs. 4.5, 4.6).
A model (n = 42) is represented in figure 4.4 that, because of its particular spacing, simulates a deep excavation. The effect of the redistribution of the stresses has clearly shown (fig.4.7 and 4.8) by the trend of the σ_y stresses in the blocks and by the curves (in black) overlaying the layer planes relative to the normal and tangential stresses on these planes. The contemporary presence of *ground* and *roof arch* effects reduces the loads that are more directly acting on the excavation only on the easily identifiable *"border effect"* area (H/T). On its inside, the normal stresses at the abutment and midspan joints increase rapidly till reaching a value that is 6 times higher than the characteristic ones of the zone subjected to *ground-arching*; the stresses normal and tangential to the layer planes tend again to assume null values for almost all of the their extension, limiting the interaction between the layers only to the extremity blocks. In the model, the *ground-arch effect* determines the transfer of the loads at the sides of the model and gives rise to a central zone, close to the roof of the void, which is subject to modest tensile stresses (fig. 4.7) which, in turn, determine the absence of stresses normal to the layer planes and therefore the detachment of the beams that characterises the *"border effect"* zone.
In order to evaluate the beneficial effect offered to the stability conditions by the presence of both the arch effects, models have been developed that are made up of 14 beams (shallow excavation) and of 70 beams (deep excavation), for different values of the vertical discontinuity spacings (a/T =2, 1 and 0.6). From the comparison (fig. 4.9) it emerges that the normal stress at the abutment joint, in the deep model, reduce by 25% and the *maximum normal stress* at the most external fibre of the roof beam in the same section is always lower (20% on average). Finally, the *redistribu-*

tion of the stresses determines, in all cases, a reduction of the entity of the values of the *displacements along the vertical* for the middle; in particular for the deepest layer (roof), the maximum differences occur in the case of lower spacings (a/T = 0.6), with values equal to about 22%; these values diminish for greater spacings up to values around 7% in the case of the a/T = 2 m ratio. This phenomenon can be explained by the increase in the extension of the zone subjected to border effects with the higher spacing (fig. 4.2).
(The parts in blue represent the trend of the stresses normal to the side and middle joints. The distribution of the principal stresses is shown on the right hand side. The final deformed configuration (amplified 50 times) is shown in the right hand side.)

5. Comparison between simplified models and tunnel models in stratified rock masses.

An attempt is now made to evaluate the approximation degree of the behaviour of tunnels in stratified rock masses from the simplified introduced models. A tunnel with a square face section of 12 m and an overburden of 56 m in a sub-horizontal stratified rock mass is considered with a mean spacing equal to 1 m and subject to persistent discontinuities orthogonal to these spacings.
The models of the behaviour of the *rock material* and the *joints* are the same as in the previous analyses. The mechanical properties that were adopted are given in table 1. Comparisons are made trough simplified multilayered models with extension equal to that of the tunnel overburden (fig. 5.1).
The results that have been obtained validate the elementary models that were used in the study; their capacity to deal with the stress and strain aspects in the most significant zones for the stability is in fact good (fig. 5.2).
The validity of the models is confirmed both by the correspondence found for the static equilibrium mechanisms and for the extension of the areas that suffer from substantial border effects (fig. 5.1) and by the distribution of the normal stresses in the critical sections of the roof beam. In these zones the maximum differences do not exceed 20%. However, the presence of side blocks, introduced to delimit the simple models, determines some slight differences, in terms of the deformation trend of the layers close to the vertical joint for the side wall.
These differences in behaviour determine, in the zones furthest from the excavation and therefore less critical for the stability, differences in the distribution of the normal stresses and in the vertical displacements along the middle. These latter also suffer from the modelling method and, more particularly, from the contemporary application of gravity and excavation of the void which implicitly occur in elementary models. The correspondence between the assumed form of the roof of the tunnel and of the lower beam of the group in the elementary models is satisfactory.

6. Conclusions

The distinct element study of the stability of underground excavations in stratified rock masses has identified the main static mechanisms that are at the basis of the equilibrium.
These mechanisms lead the stability of the excavation, independently of its depth, to that of the roof-layer and validate the approach to the problem with voussoir-beam model.
The analyses that were carried out have led to some significant results:

- The introduction of the longitudinal stiffness parameter has brought the dependence of the stability conditions to only 4 parameters, with remarkable benefits in terms of quickness in the numerical study and comparison with analytical and numerical methods.
- The analyses of the confinement-friction-fracture state connection has led to the definition of the stability and instability fields of the beams and to identify a limit value of the critical friction angle of the joints, below which it is not possible to encounter stability conditions.
- In imminent collapse conditions due to buckling, the reacting section of the beam reaches a mean value lower than 35% and the maximum deflection is equal to 22% of its thickness. These values, which are close to those supplied by the iterative method of D. & K., validate the method as a design instrument that is able to supply conservative type results.

The applicability of the voussoir beam model has been further validated by a study on multilayered beams systems which better represent the real behaviour of stratified rock masses. From this study it has been deduced that the value of the maximum deflection of the roof layer however results to be lower than that relative to the case of a single voussoir beam, and that the maximum longitudinal stresses in the critical sections (in particular at the abutment) of the single beam result to be higher than the analogous ones of the roof beam of the stratified models. These observations would justify the approach to the stability of deep underground excavations with the simple voussoir beam model.
Comparisons between simplified models and tunnel models validate the applicability of the design criteria and the theories proposed in literature for the designing of excavations in stratified and fractured undergrounds, in that they are cautionary due to the impossibility of considering all the stresses that in fact are at stake: additional friction resistance that develops on the upper face of the blocks, inter-layer friction and the existence of rock bridges.

7. References

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